Where Should BU's BigBelly Bins Really Be?

A Data-Driven Case for Smarter Waste Management

Team: Bin There Done That

Boston University · Sustainability Data Initiative

BOSTON UNIVERSITY

Our Team



Tasnim Ahmed

MS Urban Informatics
Northeastern University



Thaer Burgan

BS Data Analytics
Bentley University



Lauren Chagaris

MS Data Science
Boston University



Bidipta Roy

BS Computer Science
Boston University



Aisha Naseema

MS Applied Mathematics
Northeastern University

Problem Statement

BU Sustainability

Zero Waste Plan: Diverting 90% of waste from landfills by 2030

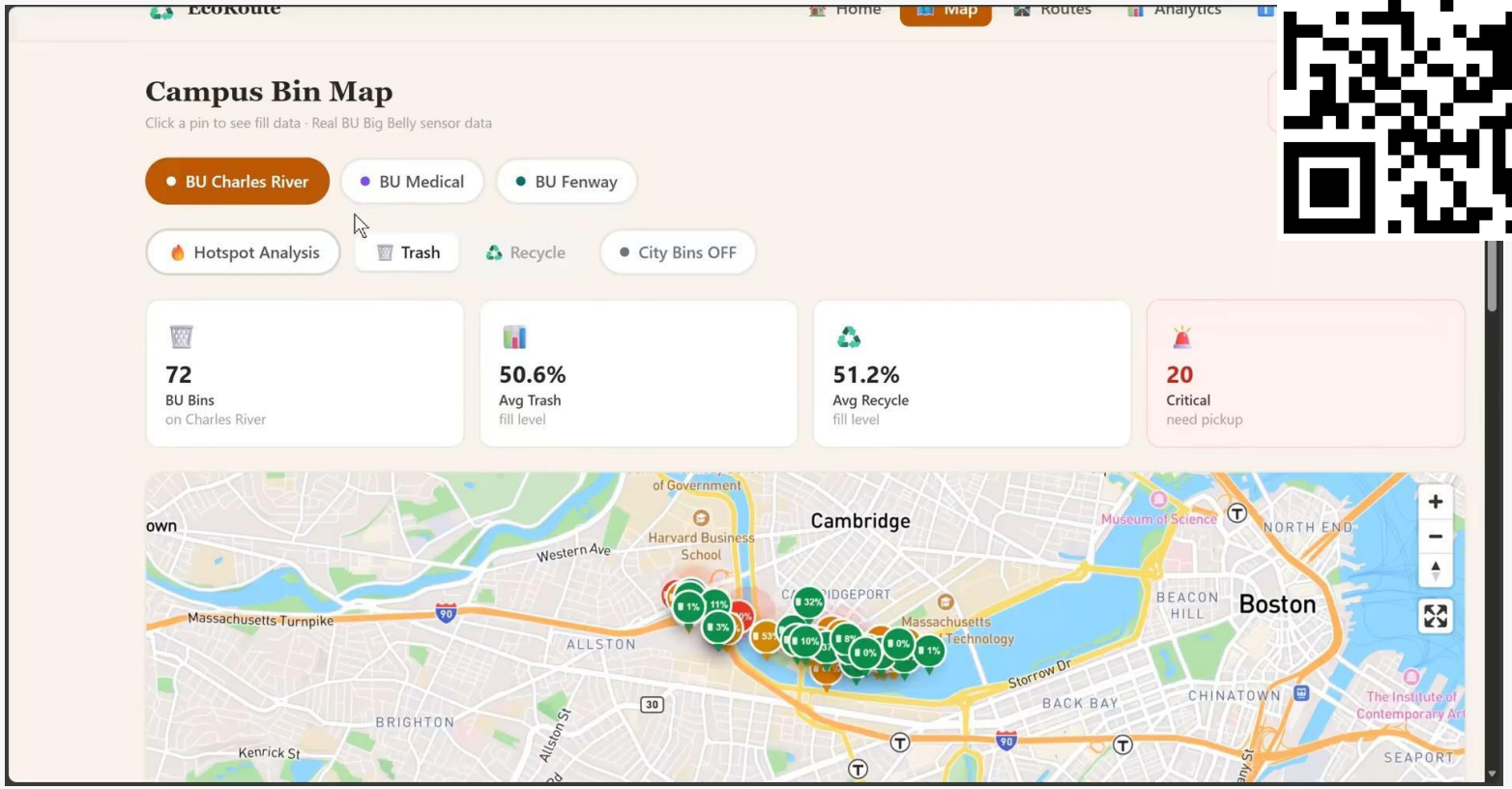


BigBelly smart bins across campus



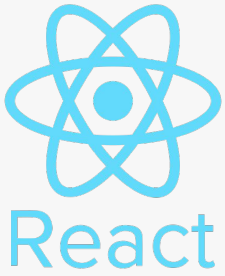
Bins overfill in high-traffic locations, bins are underutilized in low-traffic areas

Recorded Demo



https://bin-there-done-that-green.vercel.app/

Technical Implementation AND SQL LITE!!!







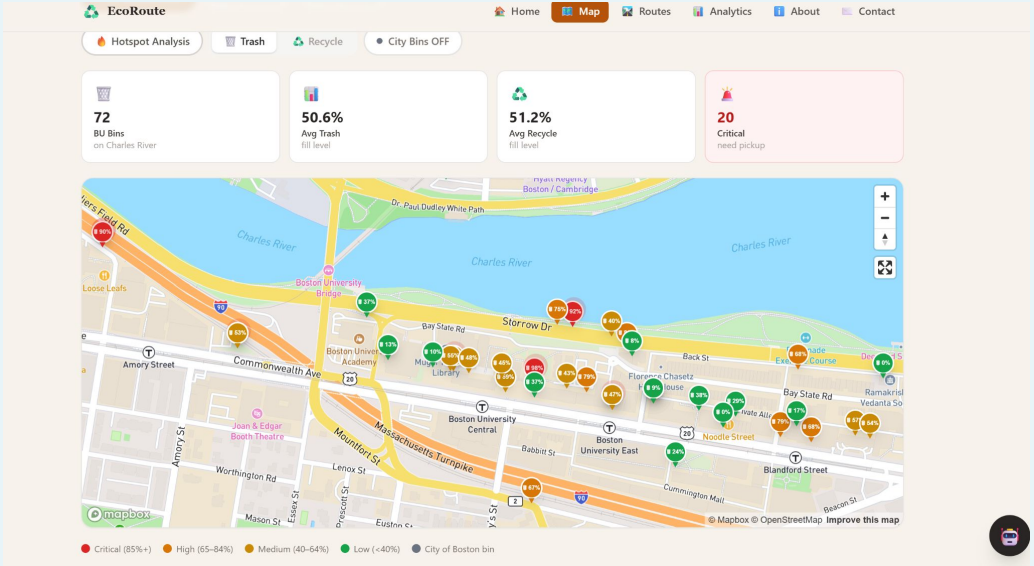




Key Findings

BU-owned vs. City-owned bins on Commonwealth Ave

COMMONWEALTH AVE CORRIDOR · BU Bridge ⟶ West Campus



Key Finding

BU should move bins in coldspots to hotspot areas

Optimized Routes
Allow for less frequent waste pickup in these areas

BU should acquire the City of Boston bins in the Hotspot areas

Ethics & Responsibility

Responsible by design

Privacy & Fairness — Only bin-level location and fill data is collected. No personal information. Benefits all Comm Ave pedestrians equally.

Transparency — Thresholds, methodology, and dashboard access are fully open to the BU Sustainability team.

Human Oversight — Data informs decisions; people make them. The system recommends, operations staff decides.

Thank you!

Questions?



Bin There Done That